

1.10 Segment Bisectors

Lesson Introduction

 Benchmark	MA.912.GR.5.2 Construct the bisector of a segment or an angle, including the perpendicular bisector of a line segment.		 Learning Targets	- I can construct the bisector of a segment. - I use the definition of a midpoint to solve problems.	
 Benchmark Coherence	Building On			Working Toward	
	N/A			N/A	
 Essential Questions	- Where is a segment bisector located? - How can the midpoint be determined? - How can knowing midpoints help solve for variables?				
 Lesson Narrative	Through exploration with the use of patty paper, students determine the relationship between the midpoint of a segment and a segment bisector while also making connections to transformations. Using this relationship, students will solve for unknown variables and find the length of line segment.				
 Component Summary	1.10.1 Warm-Up (~5 min.)	Notice and wonder about longest pedestrian bridge (<i>SE p. 64</i>)	1.10.4 Guided Practice (10 min.)	Practice using segment bisectors to solve for segment lengths (<i>SE p. 67</i>)	
	1.10.2 Exploration (15 min.)	Exploring midpoint and segment bisectors (<i>SE p. 64</i>)	1.10.5 Your Turn! (5 min.)	Independent practice using segment bisectors to solve for segment lengths (<i>SE p. 68</i>)	
	1.10.3 Guided Instruction (10 min.)	Discover using segment bisectors to solve for segment lengths (<i>SE p. 66</i>)	1.10.6 Wrap-Up (~5 min.)	Error Analysis of solving for segment lengths (<i>SE p. 68</i>)	
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> - Midpoint - Segment Bisector <u>Additional Terms:</u> N/A		- Patty Paper - Ruler (Optional) Geoboards (Optional) Tape or Glue		1.10.1 Warm-Up contains a video to accompany the question prompts.

 Teacher Tips	Common Misconceptions	Things to Consider	
	- Students may think a segment bisector must intersect at a right angle. - Students may struggle setting up the equations for solving for the line segment lengths.	- Always mark the given information on the diagram so that students can see the relationships between the lengths of the different line segments. - Consider using information from previous lessons to determine which students could use help in solving equations. Work with these students during 1.10.4 Guided Practice and 1.10.5 Your Turn! while releasing students who can solve equations to work individually. - If time is an issue, consider assigning 1.10.5 Your Turn! for homework.	
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard	
	Notice and Wonder Students will explain what they notice about the image of the bridge provided in 1.10.1 Warm-Up. This routine provides a safe place for students to brainstorm and to consider objects without any fear of being wrong. Encourage students to notice beyond the obvious. Instill in your students the idea of wonder. Consider helping students frame their wondering by asking questions, such as: - What do you think an 8-year-old would wonder? - What do you think your parent/guardian would wonder? - What do you think I would wonder? Removing the pressure of being right may help free students to imagine.	MA.K12.MTR.5.1: Patterns and Structure Throughout the lesson, students will use details from given information to solve for segment lengths. By connecting the concepts of midpoint and bisector, the students will use and mark images to set up the relationships needed to write the equations used for determining the segment lengths.	
 Differentiate Instruction	For 1.10.1 Warm-Up question 2, using Geoboards, give students an opportunity to explore other shapes they can make with overlapping straight lines. This kinesthetic activity may encourage students to become engaged with the course. An alternative to a Geoboard could be a square piece of wood with posts lining the edges. Students can tie strings to the posts and recreate this same image.		
Lesson Notes:			

1.10.1 Warm-Up: Notice and Wonder

Component Overview: Students begin this lesson by looking at an image of the longest pedestrian footbridge. Students state what they notice and wonder.



1.10.1 Warm-Up: Notice and Wonder

In May 2021, the world's longest suspended pedestrian bridge opened in Arouca, Portugal. The bridge runs over 1700 feet across a canyon, nearly 600 feet above the Paiva River.



1. What do you notice?

Answers will vary. I notice that there are a lot of geometric figures used to create the bridge, including hexagons and triangles. I notice the bridge sags in the middle.

2. What do you wonder?

Answers will vary. I wonder if the bridge sways back and forth in the wind. I wonder how long it took to build the bridge. I wonder how they were able to construct the bridge over the canyon.



Notice and Wonder

Students will have many things they notice. Encourage students to take risks and let their creativity explore options.



Students who are particularly intrigued by the bridge could further explore the engineering of the bridge.

1.10.2 Exploration: Segment Bisector

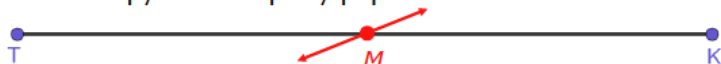
Component Overview: Using patty paper, students will identify the midpoint and draw a segment bisector. Through discussion, students will determine which transformations will map half of a line segment onto another. Using the definition, students draw different types of segment bisectors and the point of intersection is defined as the midpoint.



1.10.2 Exploration: Segment Bisector

1. Complete the following.

a. Copy $\overline{TK}$ onto patty paper.



b. Follow the steps on the copy of $\overline{TK}$.

- Fold the copy of $\overline{TK}$ that is on patty paper so the endpoints T and K align.
- Place point M on $\overline{TK}$ where the paper folds.
- Copy point M onto $\overline{TK}$ above.
- Draw a line, a line segment, or a ray through point M .

c. Discuss with your partner what you notice about $\overline{TM}$ and $\overline{MK}$.

Answers will vary. They are the same length.

The **midpoint** of a segment divides it into two congruent segments.

2. $\overline{WX}$ is shown where point P is the midpoint.



a. Discuss with your partner what transformations can be used to show that $\overline{WP}$ maps to $\overline{PX}$.

Answers will vary. Transformations that can show that $\overline{WP}$ maps to $\overline{PX}$ are

- a rotation 180° at point P .*
- a reflection over point P .*
- a translation of $\overline{WP}$ to the right.*



Students may or may not draw the segment bisector intersecting the line segment at a right angle. As long as it is drawn through the midpoint, any angle with the line segment is acceptable.

Consider allowing students to tape or paste their patty paper into their workbook.



This is a discussion opportunity. Students do not need to write an explanation here, but a sample response is provided for reference. Listen as students discuss their thinking in response to this prompt in question 2a and what students share in question 2b to determine student thinking to highlight or misconceptions to address.

1.10.2 Exploration: Segment Bisector (continued)

- b. Ask two classmates which transformations they selected. Record their transformation and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

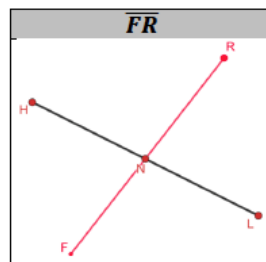
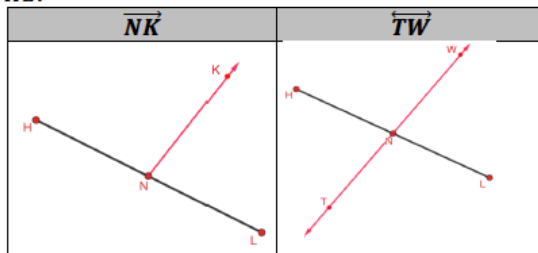
Answers will vary.

Transformation	My Summary of Their Reason	Initials

A **segment bisector** is a line, ray, or line segment that cuts a line segment into two equal parts.

3. $\overline{HL}$ is shown where point N is the midpoint.

- a. Use each description to draw a segment bisector for $\overline{HL}$.



Having students listen to their classmates and then write down what the classmate said helps the student think through their own response. Students may hear different ways to express the same thing, thus building their own mathematical thinking.



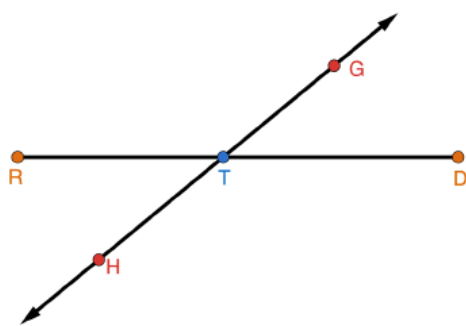
Segment bisectors are not required to intersect the line segment at right angles. Any angle is acceptable.

1.10.2 Exploration: Segment Bisector (continued)

- b. Compare your drawings with your partner.
- c. Using the definition of a segment bisector, discuss with your partner if all of the segment bisectors have to be at the same angle with $\overline{HL}$. Summarize your discussion.

Answers will vary. No, a bisector only has to cross in the middle, the angle it crosses is not important.

4. $\overline{RD}$ and $\overline{HG}$ are shown where point T is the point of intersection.



- a. Use patty paper to copy $\overline{RD}$.
- b. Fold the copy of $\overline{RD}$ to determine if point T is a midpoint.
- c. Complete the sentence.

$\overline{HG}$ ☐ is ☒ is not a segment bisector because point T

☐ is ☒ is not a midpoint.



Students may incorrectly fold their paper along line HG instead of folding at point T so that $\overline{RT}$ coincides with $\overline{TD}$. If you notice this happening, instead of correcting them, ask questions about which line they are trying to bisect, which segments they are trying to determine, and if they have equal lengths.

1.10.3 Guided Instruction: Determining Lengths

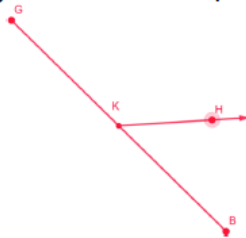
Component Overview: Using the definition of a segment bisector, solve for unknown variables and line segment lengths.



1.10.3 Guided Instruction: Determining Lengths

1. $\overrightarrow{KH}$ bisects $\overline{GB}$ at point K .

- a. Draw a diagram of the description.

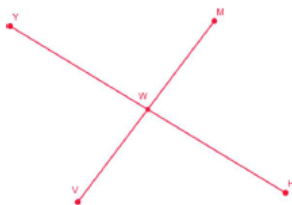


- b. Determine the value of x if $GK = 7x + 36$ and $KB = 13x - 18$.
 $x = 9$

- c. What is the length of $\overline{GB}$?
 $GB = 198$

2. $\overline{YH}$ and $\overline{MV}$ bisect each other at point W .

- a. Draw a diagram of the description.



- b. Determine the value of b if $YH = 12b - 4.5$, $MV = 17b + 0.75$, and $MW - WH = 13.25$.
 $b = 4.25$



Take care to ensure that students understand that the angle at which ray KH intersects line segment GB does not have to be 90 degrees. In fact, the preciseness of creating such an angle requires its own construction which will be addressed in unit 2.



While demonstrating question 2b, mark the diagram with the lengths that are given and can be determined from the problem. While doing this, consider asking the following questions:

- How is the length of $\overline{YW}$ related to the length of $\overline{YH}$? How can I represent this algebraically?
- How are the lengths of $\overline{YW}$ and $\overline{WH}$ related?
- How is the length of $\overline{VW}$ related to the length of $\overline{VM}$? How can I represent this algebraically?
- How are the lengths of $\overline{VW}$ and $\overline{WM}$ related?

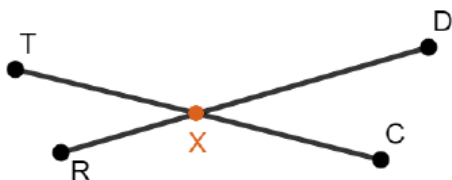
1.10.4 Guided Practice: Determining Lengths

Component Overview: Using the definition of a segment bisector, solve for unknown variables and line segment lengths.



1.10.4 Guided Practice: Determining Lengths

1. $\overline{RD}$ bisects $\overline{TC}$ at point X .



- a. Determine the value of x if $TX = \frac{16}{3}x - 41$ and $XC = 3x + 22$.
 $x = 27$
- b. What is the length of $\overline{TC}$?
 $TC = 206$



Consider pairing students together for this question and naming them Partner 1 and Partner 2. For question 1a, have Partner 1 do all of the talking while Partner 2 does not talk, and only writes what Partner 1 says. When Partner 1 is finished talking, Partner 2 can agree with the work or highlight in the work where an error occurred.

For question 1b, students swap roles.

1.10.5 Your Turn!

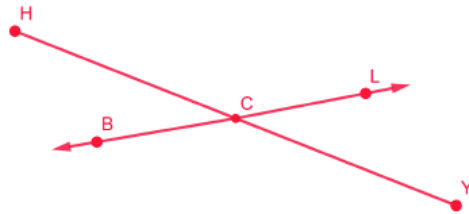
Component Overview: Students draw a diagram of the description, solve for the variable, and determine the length of a line segment.



1.10.5 Your Turn!

1. $\overleftrightarrow{BL}$ bisects $\overline{HY}$ at point C .

- a. Draw a diagram of the description.



- b. Determine the value of x if $HC = 5x + 9$ and $CY = 8x - 36$.
 $x = 15$

- c. What is the length of $\overline{HY}$?
 $HY = 168$



Consider asking these follow-up questions:

- Why is it important that we are asked to find the midpoint of the line segment rather than the line?
- Why might it be impossible to find the midpoint, or draw a bisector of a line, but it is possible on a line segment?

1.10.6 Wrap-Up: Error Analysis

Component Overview: Students will read through the work of other students and write notes to the students indicating their mistakes.

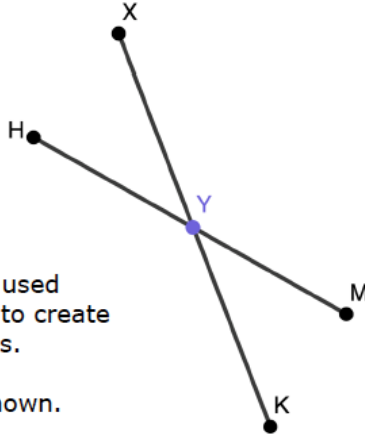


1.10.6 Wrap-Up: Error Analysis

- $\overline{XK}$ and $\overline{HM}$ bisect each other at point Y , where $XY = 5x + 3$, $HY = 2y + 5$, $XK = 11x + 2$, and $YM = 3y - 2$.

Taliyah and Giancarlo used the given information to create two different equations.

Their equations are shown.



Taliyah	Giancarlo
$5x + 3 = 2y + 5$	$5x + 3 = 2(11x + 2)$

Both Taliyah and Giancarlo made a mistake. If Taliyah was trying to find the value of y and Giancarlo was trying to find the value of x , determine each of their errors. Explain each error.

Taliyah	Giancarlo
Taliyah set the wrong line segments equal to each other. She should have set $HY = YM$ to create the equation $2y + 5 = 3y - 2$.	Giancarlo doubled the length of the full line segment, $\overline{XK}$, when he should have taken half the length and set equal to XY .



This question assesses student understanding of using segment bisectors to solve problems. Students' responses to this activity can provide data on which students may need additional support with this content.